TOPIC 5/60

What is a Transformer?

> The "T" in ChatGPT.
>
> The architecture that ignited the AI revolution.

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THE ORIGIN STORY

In 2017, a team of researchers at Google published a paper that completely altered the trajectory of AI history.

PAPER TITLE:

"Attention Is All You Need"

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THE PARADIGM SHIFT

Breaking the Loop

Before Transformers, AI read text like a human: sequentially, one word at a time (RNNs).

The Breakthrough: The Transformer discards the sequential loop entirely. It reads the entire sentence (or book) **simultaneously at the exact same time**.

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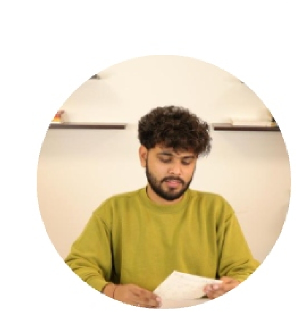

KEY MECHANISM 1

Self-Attention

Context is Everything

Because it looks at all words at once, it calculates how much each word relates to every other word in the sentence.

Example:

1. "I went to the **bank** to deposit money."
2. "I sat on the **bank** of the river."

Self-attention allows the AI to instantly realize "bank" has two completely different meanings based on the surrounding context.

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KEY MECHANISM 2

Positional Encoding

Keeping the Order

If the AI processes every word simultaneously, how does it know the order of the sentence?

> **The Solution:** It injects a mathematical "timestamp" (positional encoding) into every word. This allows the AI to perfectly reconstruct the sequence without reading it sequentially.

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THE BIG ADVANTAGE

Parallelization

Unleashing the GPU

Because Transformers don't wait for word #1 to finish before processing word #2, they can be parallelized.

The Speed Run:

You can split the training of a Transformer across 10,000 GPUs at the exact same time.

> This is what allowed us to train models on the entire internet.

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THE TWO HALVES

THE ENCODER (READS)

Focuses on deeply understanding the input text.

> Used in Google's **BERT** for search.

THE DECODER (WRITES)

Focuses on generating the next word perfectly.

> Used in OpenAI's **GPT**.

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BEYOND LANGUAGE

The Universal Engine

Transformers were built for translation, but they turned out to be the ultimate pattern-matching engine for *everything*.

Today, Transformers are used to generate images (Vision Transformers/ViT), analyze DNA, predict weather, and synthesize human voices.

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THE GLOBAL IMPACT

Massive Scaling

Transformers scale predictably. More data + more GPUs = a smarter model.

Foundation Models

This architecture led directly to the creation of GPT-4, Claude 3, and LLaMA.

The Trillion Dollar Boom

Nvidia's explosive growth is largely because their GPUs are the perfect hardware for running Transformers.

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The Transformer

CONCEPT UNLOCKED

- ⚡ Processes all words simultaneously
- 🕸️ Self-Attention maps context
- 🧩 Positional Encoding tracks order
- 🤖 The engine behind ChatGPT

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